Sentiment Analysis Report

Dataset: Datafiniti Amazon Consumer Reviews of Amazon Products (May 2019)
Model: spaCy (en_core_web_md) + TextBlob (spacytextblob)
Total Reviews Analysed: 28,332

1. Introduction

This project performs sentiment analysis on Amazon consumer product reviews using spaCy and TextBlob. The objective is to classify customer reviews as Positive, Neutral, or Negative based on their textual content and to evaluate the overall sentiment distribution within the dataset.

2. Description of the Dataset

The dataset used in this project is the Datafiniti Amazon Consumer Reviews of Amazon Products (May 2019) dataset obtained from Kaggle.

According to Kaggle, this dataset contains over 34,000 consumer reviews for Amazon consumer electronics products such as:

- Kindle devices
- Fire TV Stick
- Echo devices
- Other Amazon-branded electronics

The dataset includes:

- Product information
- Customer star ratings
- Written review text
- Additional metadata

For this analysis, the primary feature used was:

reviews.text – the written customer review.

After removing missing values, a total of 28,332 reviews were available and used for sentiment analysis.

3. Preprocessing Steps

Before applying sentiment analysis, several preprocessing steps were performed to prepare the text data.

3.1 Removing Missing Values

All rows with missing values in the reviews.text column were removed using:

```
clean_df = df.dropna(subset=["reviews.text"])
```

In this dataset, no rows were removed, as all entries contained valid review text.

3.2 Text Cleaning

Each review was processed as follows:

- Converted to lowercase using .lower()
- Stripped of leading and trailing whitespace using .strip()
- Cleaned to remove excessive whitespace using regular expressions

This ensures consistent formatting and reduces variability caused by capitalization or spacing.

3.3 Stop Word Removal

Stop words such as “the”, “is”, and “and” were removed using spaCy’s token.is_stop attribute.

Example:

Original – These batteries are very long lasting the price is great.

After stopword removal – batteries long lasting price great.

Removing stop words helps focus on meaningful words that contribute to sentiment classification.

3.4 Sentiment Calculation

Sentiment was calculated using TextBlob’s polarity score via spaCy:

```
polarity = doc._.blob.polarity
```

Polarity scores range from:

- -1 – Very Negative
- 0 – Neutral
- +1 – Very Positive

Sentiment labels were assigned using the following thresholds:

- Positive → polarity > 0.10
- Negative → polarity < -0.10
- Neutral → otherwise

4. Evaluation of Results

4.1 Sample Predictions

Example predictions from the first five reviews:

Review Description	Polarity	Predicted Sentiment
"bad quality… missing backup spring…"	-0.45	Negative
"bulk… less expensive way…"	-0.50	Negative
"price I am happy"	0.80	Positive
"better price"	0.50	Positive
"long lasting… price great"	0.25	Positive

The model correctly identified clear positive and negative wording.

4.2 Overall Sentiment Distribution

Out of 28,332 reviews:

- Positive**:** 21,838 (~ 77%)
- Neutral**:** 5,474 (~ 19%)
- Negative**:** 1,020 (~ 4%)

The results show a strong positive skew in the dataset. This is not unusual, as satisfied customers are more likely to leave reviews.

4.3 Polarity Statistics

- Mean polarity: 0.3709
- Minimum polarity: -1.000
- Maximum polarity: 1.000
- Median polarity: 0.3867

- Standard deviation**:** 0.3078

The positive mean polarity confirms that the dataset contains predominantly favourable reviews.

The distribution shows that while extremely negative reviews exist, they represent a small proportion of the dataset.

4.4 Review Similarity

similarity comparison between the first two cleaned reviews resulted in:

Similarity score: 0.7223

This indicates that the model found them moderately similar, likely because both relate to product quality and purchasing considerations.

5. Insights into Model Strengths and Limitations

Strengths

1. Easy Implementation

The model was simple to implement using spaCy and TextBlob without requiring model training.

2. Interpretable Scores

The polarity score (-1 to +1) provides a clear and measurable sentiment intensity.

3. Effective for Explicit Sentiment

The model performed well on reviews containing clear sentiment words such as:

- “bad quality”
- “great”
- “happy”
- “better price”

4. Scalable

The system successfully processed over 28,000 reviews.

Limitations

1. Lexicon-Based Approach

TextBlob relies on predefined sentiment dictionaries and does not deeply understand context. It may struggle with:

- Sarcasm
- Irony
- Complex contextual meaning
- Technical product language

Example:

“Great, it broke after one day.”
May be misclassified as positive.

2. Negation Handling

Removing stop words may remove important negation terms such as “not”, potentially reversing the intended sentiment.

Example:

“not good” – may become “good”

3. Dataset Bias

The dataset is heavily skewed toward positive reviews (77%), which may influence perceived model performance.

4. No Domain Fine-Tuning

The model was not specifically trained on Amazon product reviews. A fine-tuned transformer model (e.g. BERT) would likely produce more accurate results.

6. Conclusion

The sentiment analysis model successfully classified 28,332 Amazon reviews into positive, neutral, and negative categories using spaCy and TextBlob.

The dataset exhibited a strong positive bias, with approximately 77% of reviews classified as positive. The average polarity score of 0.37 further supports this trend.

While the model performs well for clearly worded reviews, it has limitations in handling sarcasm, negation, and nuanced expressions. Future improvements could include:

- Fine-tuning transformer-based models

- Avoiding removal of negation stop words
- Optimizing polarity classification thresholds
- Training supervised sentiment classifiers on labeled review data